Lectures 11-12

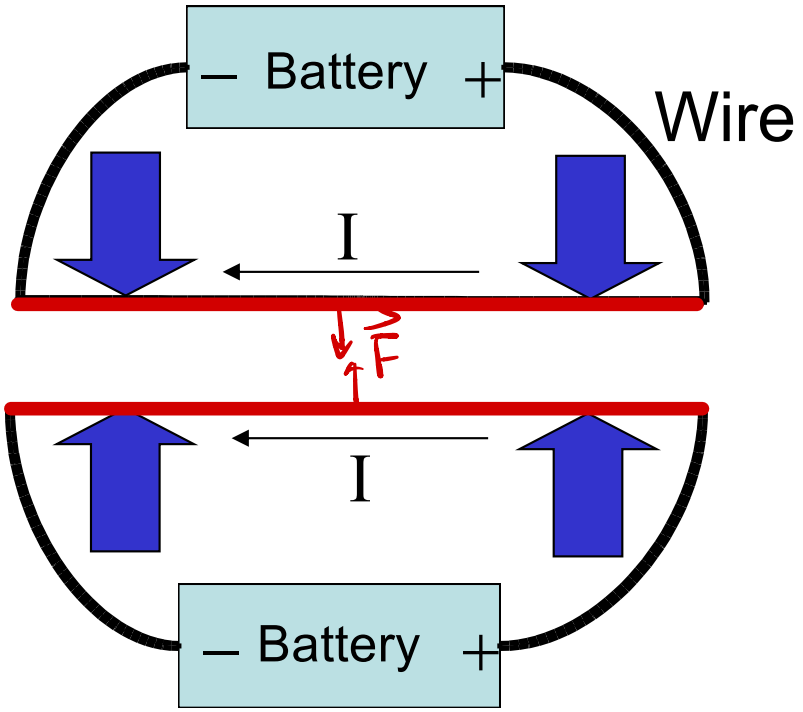
Magnetic Flux and Magnetic Fields
Biot-Savart Law
Ampere's Law

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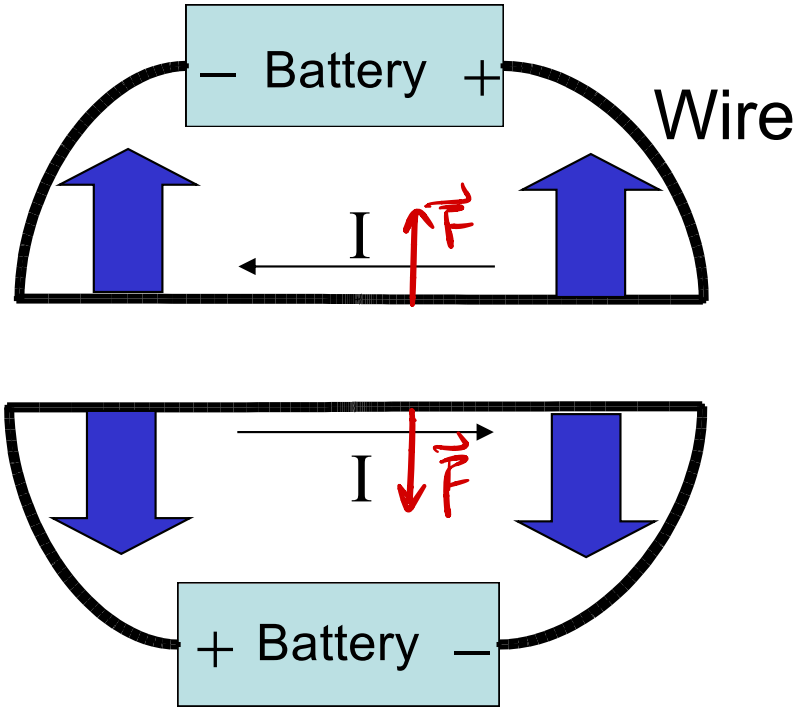
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Magnetism and Electricity

Andre Marie Ampere - 1820



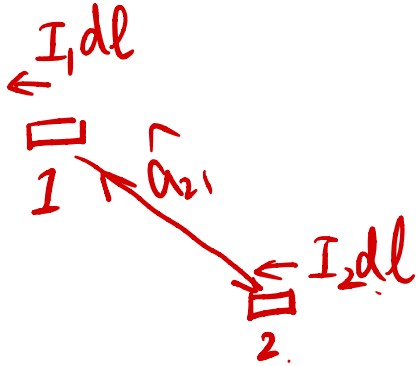
Parallel Currents in Same Direction ATTRACT



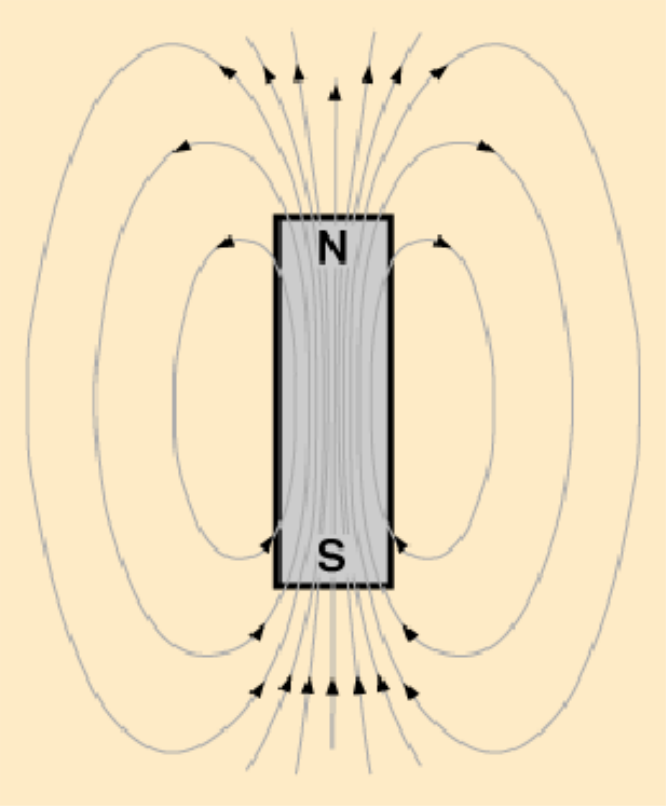
Parallel Currents in Opposite Direction REPEL

Ampere's observations

- The magnitude of F is
  - proportional to the product of the currents AND to the product of their lengths $F \propto I d\ell$
  - inversely proportional to the square of the distance $F \propto \frac{1}{R^2}$
  - depends on the medium
- The direction of F on current 1 is
  - perpendicular to $d\mathbf{l}_1$
  - perpendicular to $d\mathbf{l}_2 \times a_{21}$
- The forces $d\mathbf{F}_1$ and $d\mathbf{F}_2$ are not always equal and opposite



Magnetic Flux Lines



Magnetic flux lines form a VECTOR FIELD

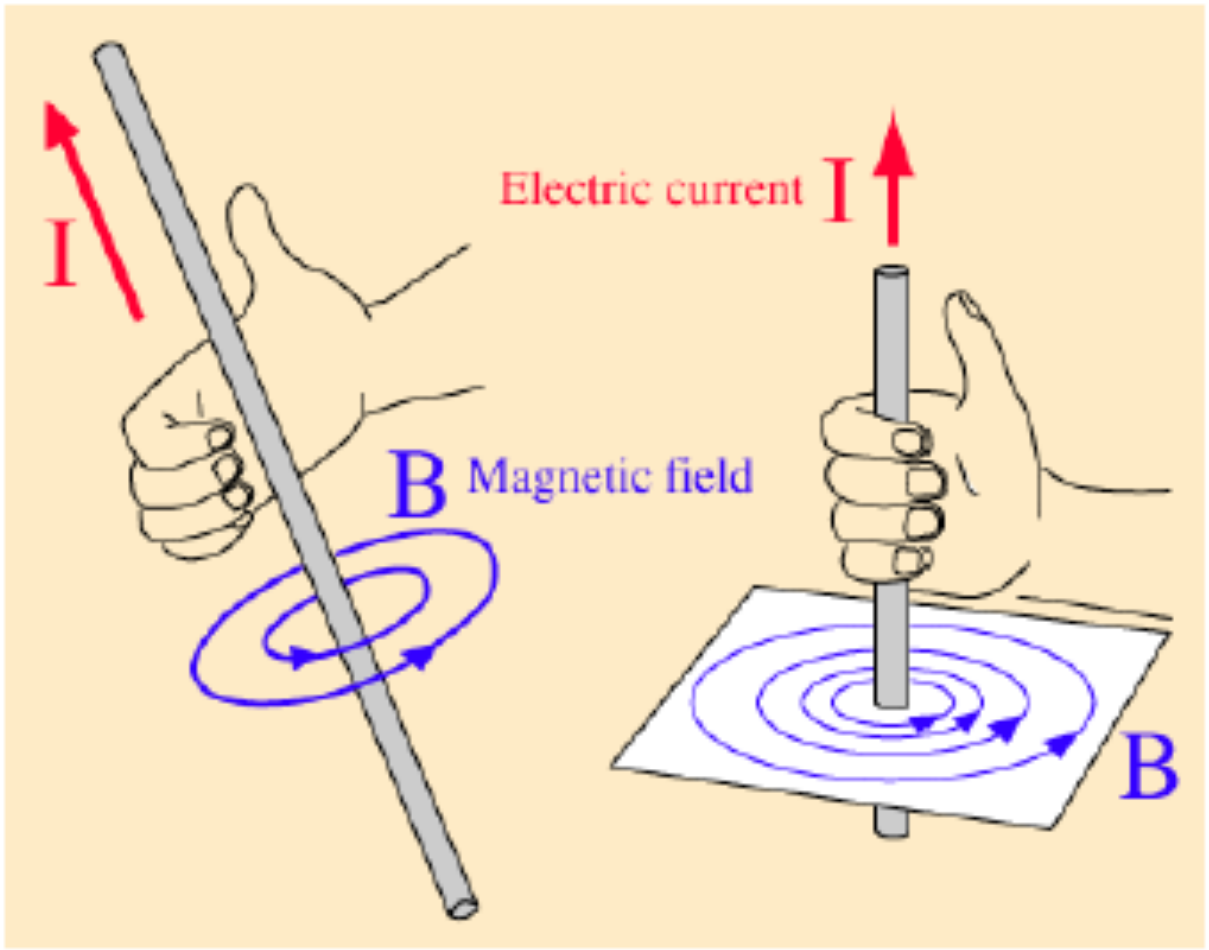
Density of lines indicates MAGNITUDE

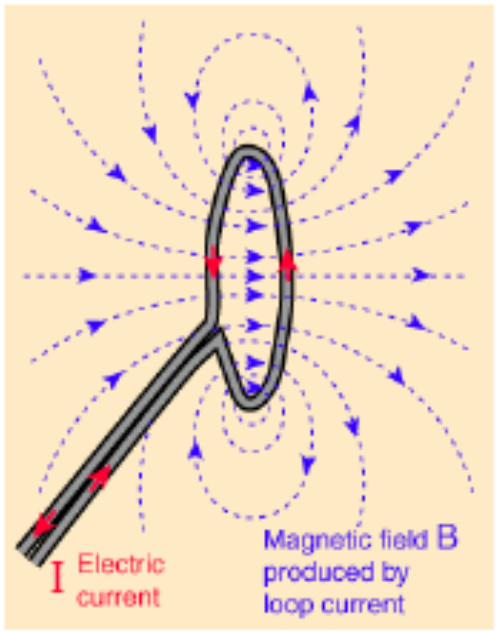
Direction = the way our compass would point

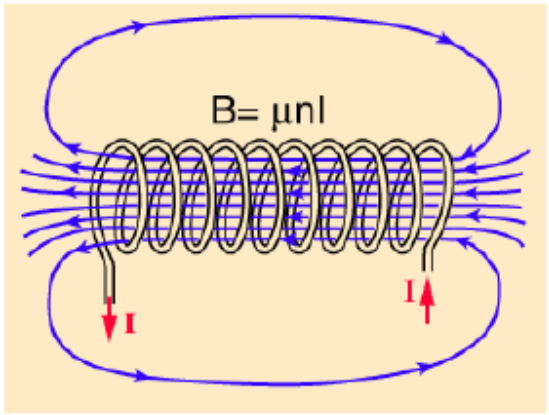
Unlike electric field lines which begin on positive charges and end on negative charges, magnetic flux lines **NEVER** begin or end

$$\oint \vec{B} \cdot d\vec{s} = 0.$$

Magnetic Flux Lines



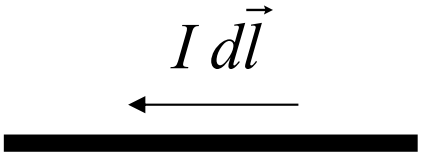




Units for **B** are Tesla = Newt/(Amp-meter) or Webers/m^2 (Magnetic flux density)

Current = Moving Charge

What is CURRENT? CHARGES IN MOTION!!



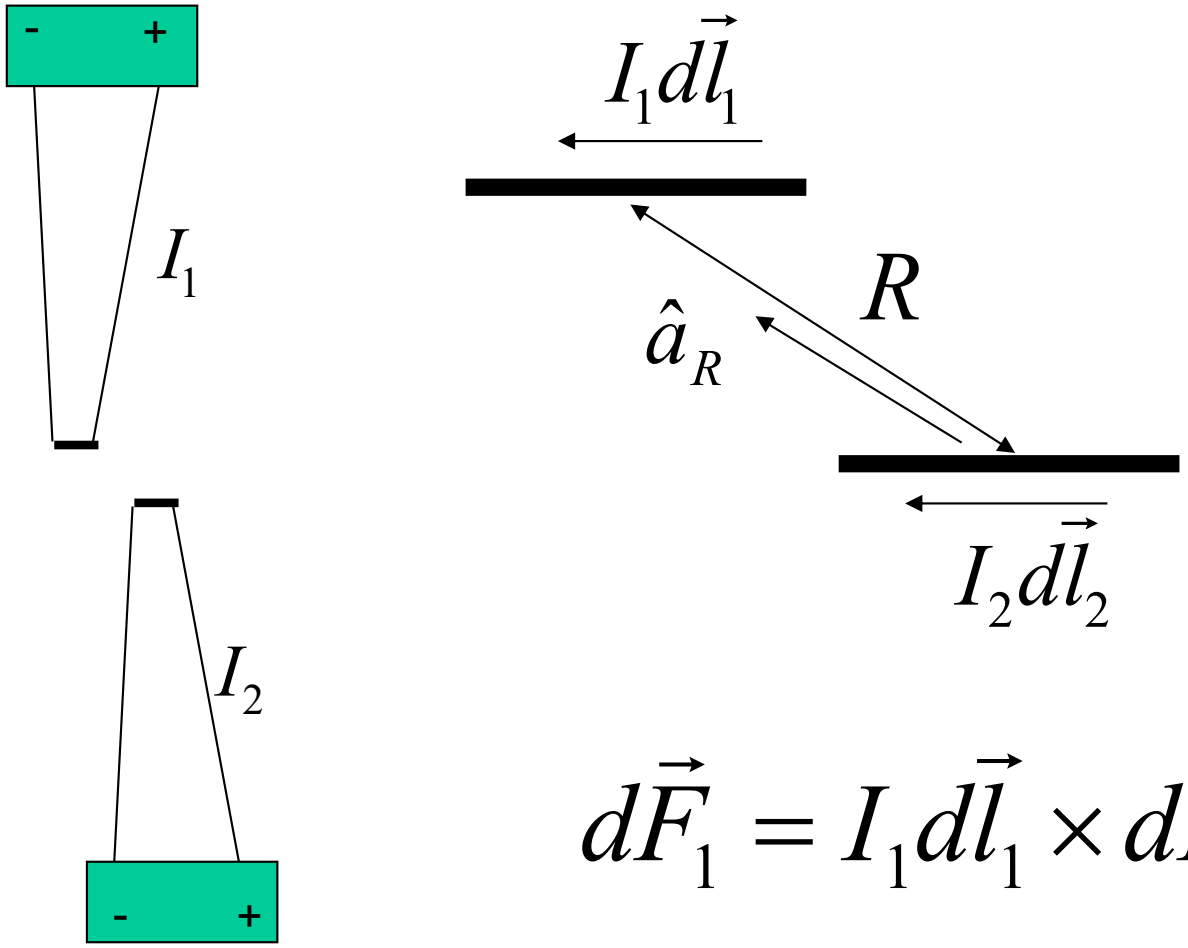
$$I\,d\vec{l} = q\,\vec{v}$$

$$\frac{coul}{\text{sec}} \cdot m = coul \cdot \frac{m}{\text{sec}}$$

So the current in a wire, ***I***, flowing across a magnetic field will feel a force…

$$F_M = (I\,d\vec{l}\,) \times \vec{B} = q\,\vec{v} \times \vec{B}$$

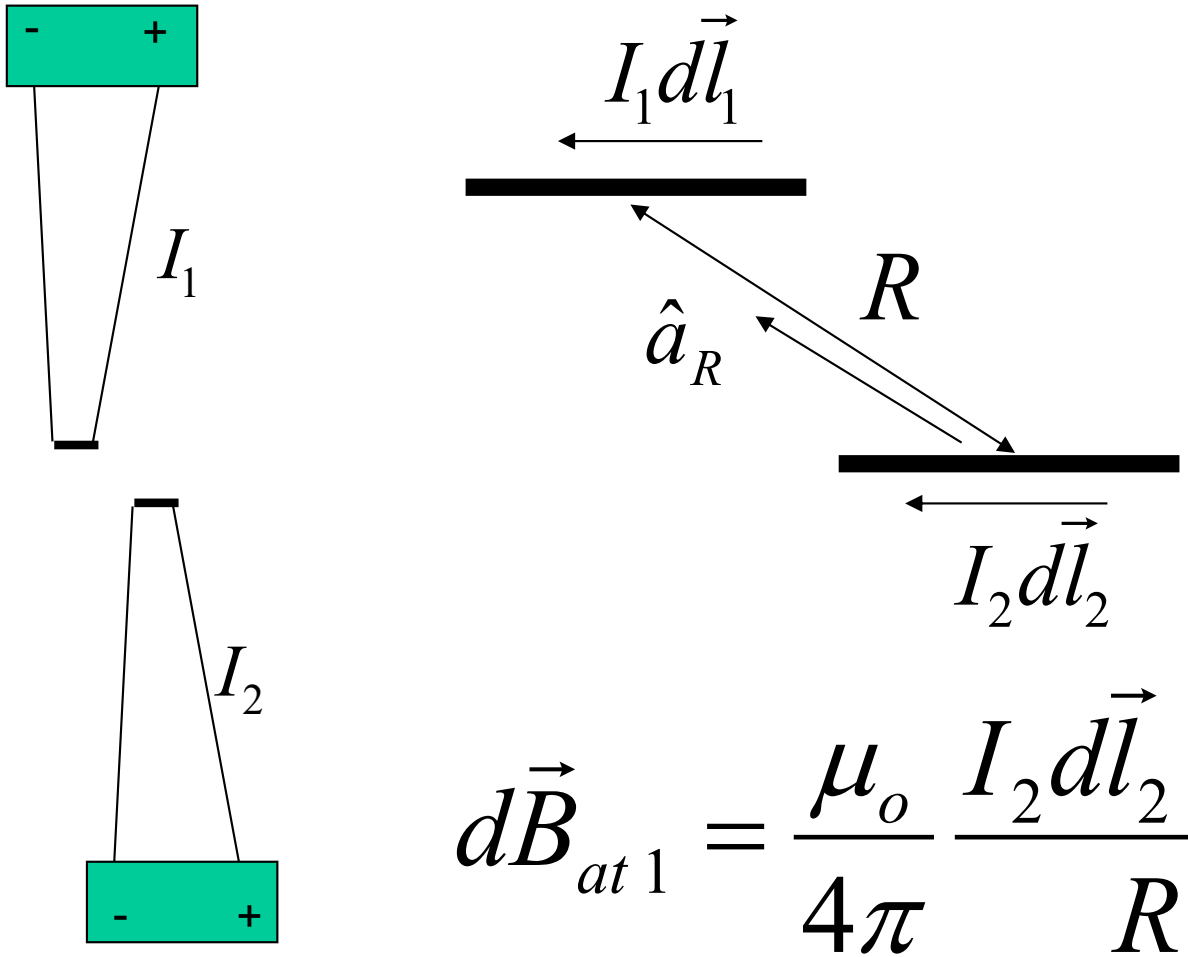
Ampere's Force Law



$$d\vec{F}_1 = I_1 d\vec{l}_1 \times d\vec{B}_1$$

Force on current 1 due to current 2 depends on the magnetic flux density at 1 due to current 2.

Biot-Savart Law for finding **B**



$$d\vec{B}_{at\,1} = \frac{\mu_o}{4\pi} \frac{I_2 d\vec{l}_2 \times \hat{a}_R}{R^2}$$

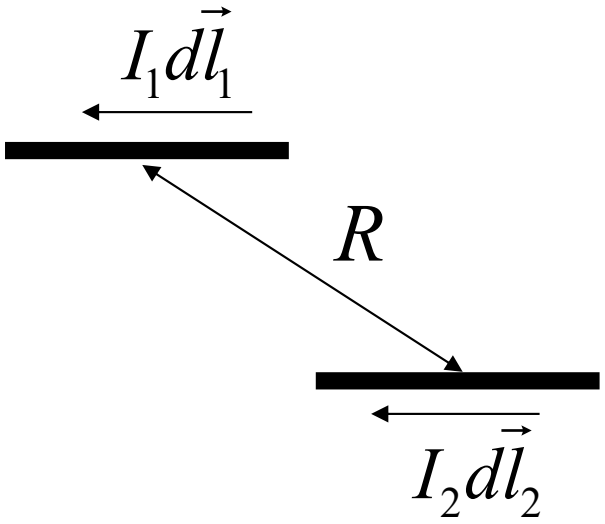
Biot-Savart law

Magnetic flux at point 1 due to current 2

Force between two wires

Combining these results,

$$dF = (I_1 d\vec{l}_1) \times \underbrace{\left(\frac{\mu_0}{4\pi} \cdot \frac{I_2 d\vec{l}_2 \times \hat{a}_R}{R^2} \right)}$$

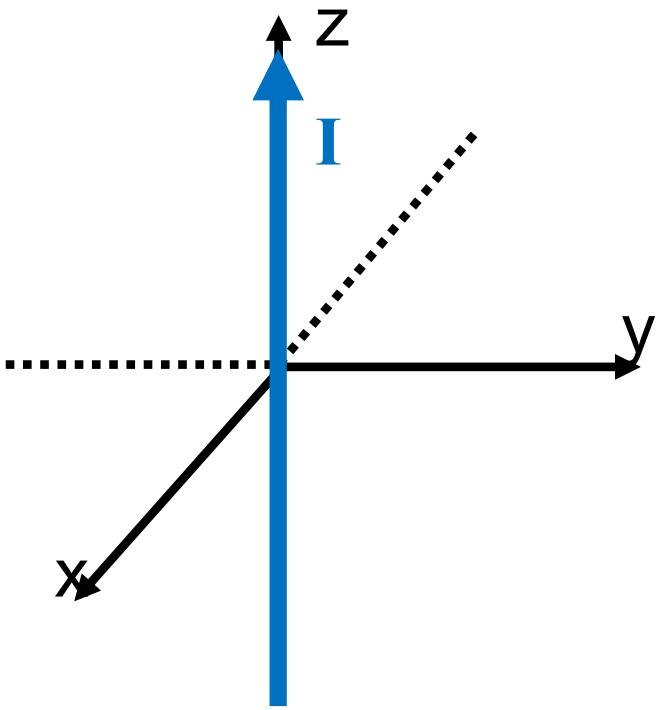


Magnetic flux density caused by #2 at #1

$$dF_{at\,1} = (I_1 d\vec{l}_1) \times \vec{B}$$

To find the total force on wire 1, we add up (integrate) the contributions from segments $d\mathbf{l}_2$ for every $d\mathbf{l}_1$ (double integral)

Example: Find **B** for an Infinite Line of Current



Wire carrying current in +z direction = $I(A)$
Find **B** for any arbitrary point $P(r, \phi, z)$ in cylindrical coordinates.

$$d\vec{B}_{at\,1} = \frac{\mu_o}{4\pi} \frac{I_2 d\vec{l}_2 \times \hat{a}_R}{R^2}$$



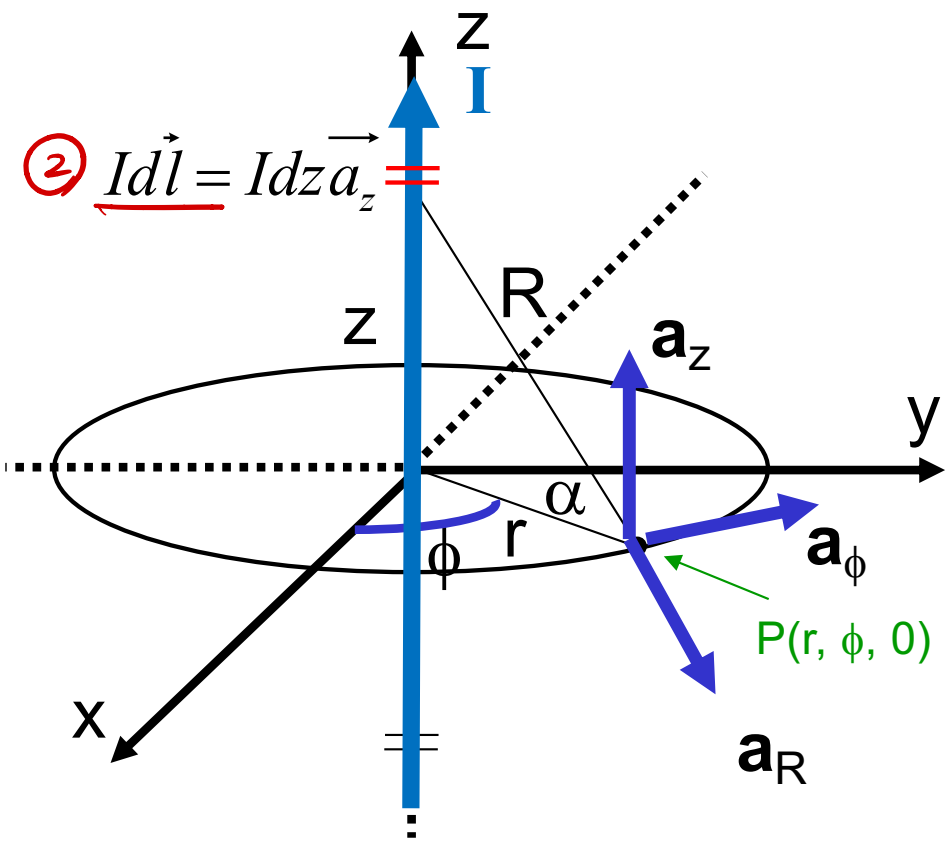
You cannot just treat an entire long wire as one segment and plug numbers into the formula.

Instead: You must break the wire into tiny elements dℓ, write the contribution dB from each, and integrate.

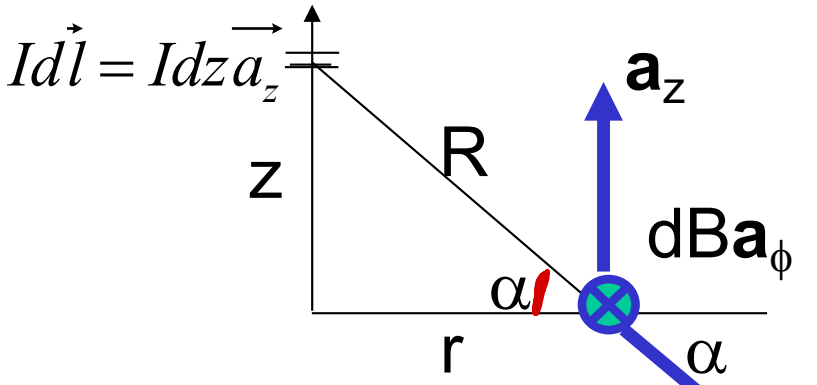
Patented 5-Step Program for Problem Solving

1. MAKE A **LARGE CLEAR** DRAWING
   a. Also draw cross-sections if the problem is in 3D
   b. Pick a coordinate system that is appropriate for the symmetry of the problem
2. Divide current distributions into tiny pieces
3. Find $d\mathbf{B}$ of one tiny piece
4. Use SYMMETRY to eliminate any components that cancel (i.e. add to ZERO)
5. INTEGRATE to add contribution of ALL the tiny pieces

Step 1 - draw a picture
Step 2 - divide the line into segments d**l**
Step 3 - find d**B** for one small segment



TANGENTIAL View



$$d\vec{B} = \frac{\mu_o}{4\pi}\frac{Id\vec{l}\times\hat{a}_R}{R^2}$$

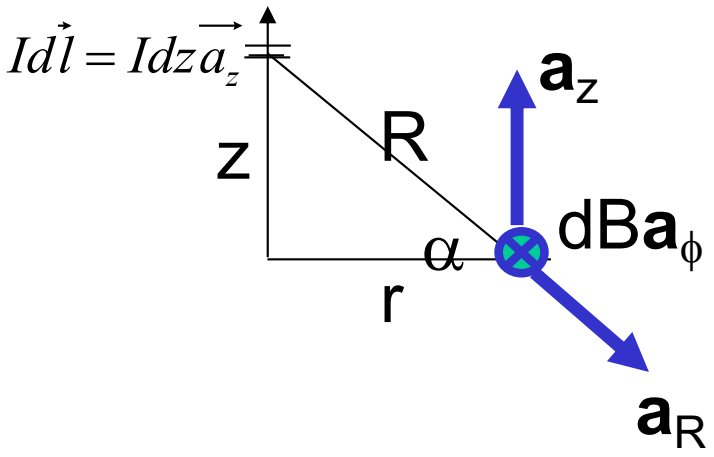
$\cos\alpha = \frac{r}{R}$

$$d\vec{B} = \frac{\mu_o}{4\pi}\frac{Idz\sin(90+\alpha)}{R^2}\hat{a}_\phi$$

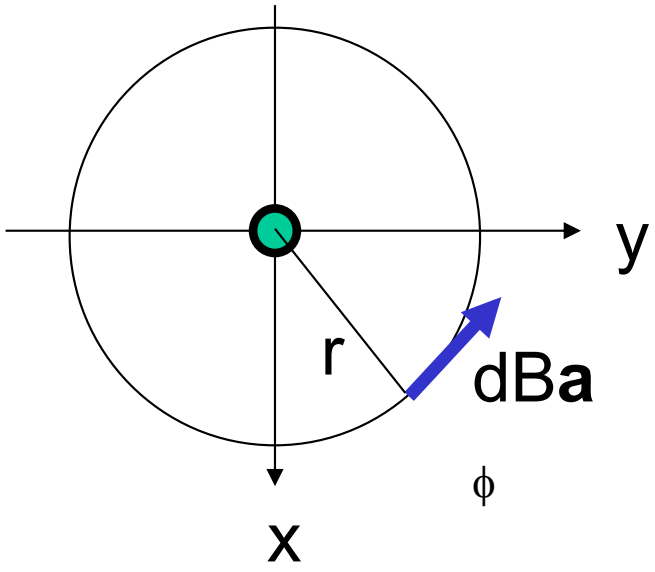
$$d\vec{B} = \frac{\mu_o}{4\pi}Idz\frac{r}{R^3}\hat{a}_\phi$$

Step 4: Use symmetry to eliminate components
Step 5: Integrate over the whole object

TANGENTIAL View



TOP View

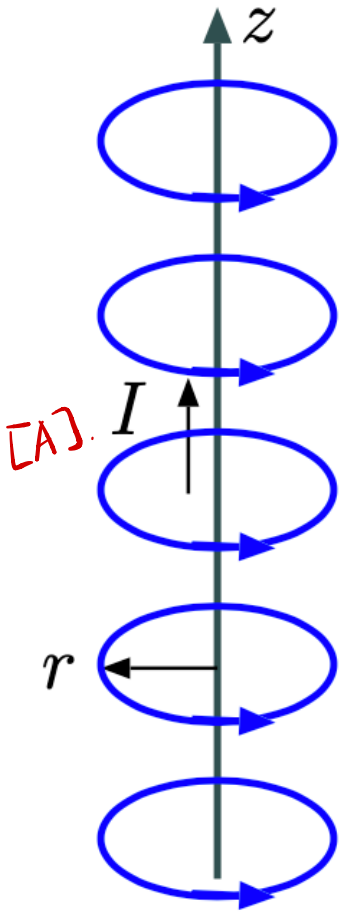


$$d\vec{B} = \frac{\mu_o}{4\pi} Idz \frac{r}{R^3} \hat{a}_\phi$$

$$\vec{\boldsymbol{B}} = \int_{-\infty}^{\infty} d\vec{\boldsymbol{B}} = \frac{\mu_o Ir}{4\pi} \hat{\boldsymbol{a}}_\phi \int_{-\infty}^{\infty} \frac{1}{(z^2+r^2)^{3/2}} dz = \frac{\mu_o Ir}{4\pi} \vec{a}_\phi \left[\frac{z}{r^2\sqrt{z^2+r^2}} \right]_{z=-\infty}^{\infty}$$

$$\boxed{\vec{\boldsymbol{B}} = \frac{\mu_o I}{2\pi r} \hat{\boldsymbol{a}}_\phi}$$

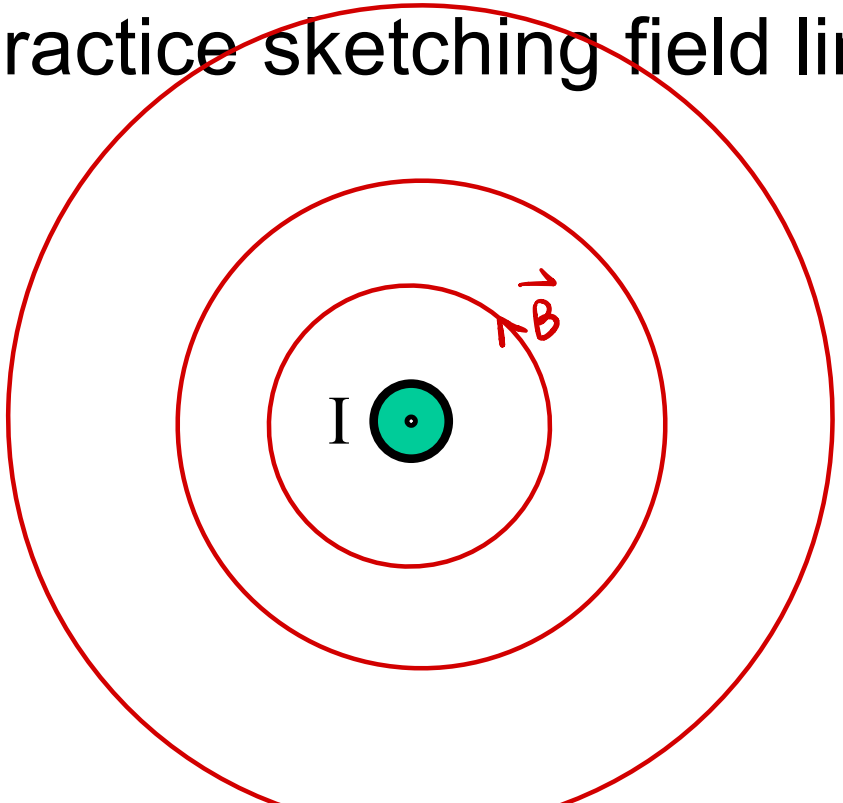
Magnetic flux density (**B**) around a line of current



$$\mathbf{B} = \frac{\mu_o I}{2\pi r}\hat{\phi}$$

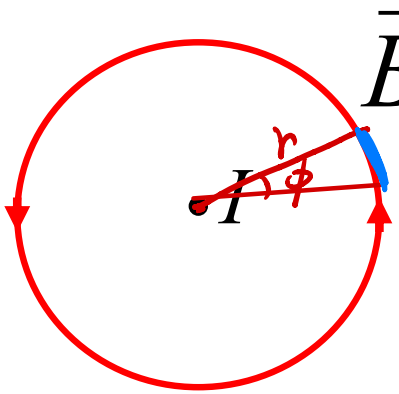
Right hand rule
Curl fingers
around current

Practice sketching field lines:



"Static" Ampere's Law

Let's integrate B over any circular path centered on the wire:



$d\vec{l} = r\,d\phi\,\hat{\phi}$

$$\oint_C \vec{B} \bullet d\vec{l} = \int_{\phi=0}^{2\pi} \left(\frac{\mu_0 I}{2\pi r} \hat{a}_\phi \right) \bullet \left(r\,d\phi \hat{a}_\phi \right)$$

$$= \int_{\phi=0}^{2\pi} \left(\frac{\mu_0 I}{2\pi} \right)$$

$B \cdot 2\pi r$

$$= \mu_0 I$$

$\oint_C \vec{B} \cdot d\vec{l} = \mu_0 I$.

[Wb/m²]

[T]

So…

$$\oint_C \frac{\vec{B}}{\mu_0} \bullet d\vec{l} = I$$

for any radius circle

$\vec{H} = \frac{\vec{B}}{\mu}$

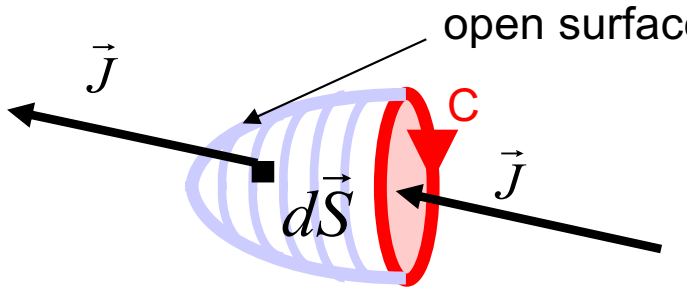
[A/m]

$\oint_C \vec{H} \cdot d\vec{l} = I$.

New Definition: Magnetic Field Intensity Vector

$\oint \vec{B} \cdot d\vec{\ell} = \mu_0 I_{enc}$

$$\oint_C \vec{\boldsymbol{H}} \cdot d\vec{\boldsymbol{l}} = I_{enclosed} = \int_S \vec{\boldsymbol{J}} \cdot d\vec{\boldsymbol{S}}$$



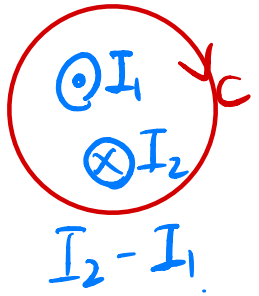
Define magnetic field

$$\vec{H} = \frac{\vec{B}}{\mu_0} \qquad \text{Units: (A/m)}$$

Use Stoke's theorem

$$\oint_C \vec{\boldsymbol{H}} \cdot d\vec{\boldsymbol{l}} = \int_S \nabla \times \vec{\boldsymbol{H}} \cdot d\vec{\boldsymbol{S}} = \int_S \vec{J} \cdot d\vec{S}$$

$[A/m^2]$.



Differential form of Ampere's law

$$\nabla \times \vec{\boldsymbol{H}} = \vec{\boldsymbol{J}}$$

$$\nabla \cdot \vec{\boldsymbol{H}} = 0$$

Laws of magnetostatics

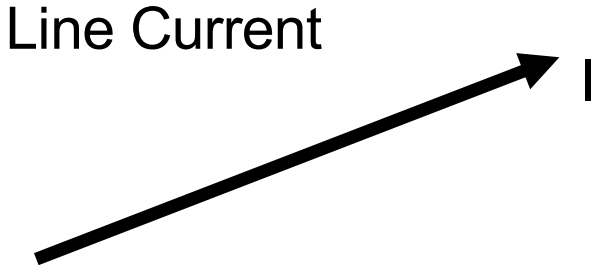
electrostatics.

$\oint \vec{B} \cdot d\vec{S} = 0 \rightarrow$

$\oint_S \vec{H} \cdot d\vec{S} = \iiint_V \nabla \cdot \vec{H}\, dV = 0$.

$\begin{cases} \nabla \times \vec{E} = 0 \\ \nabla \cdot \vec{D} = \rho \end{cases}$

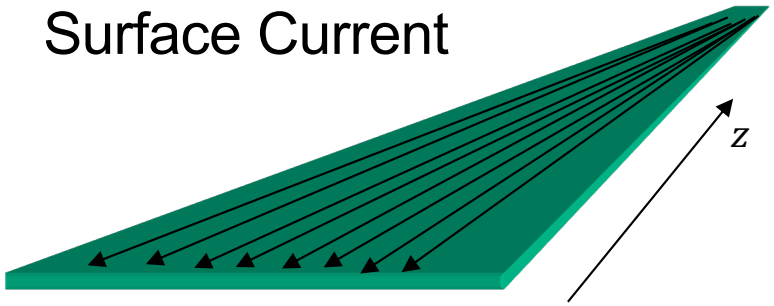
Current Distributions



line current density

$$\vec{J}(x,y,z) = \hat{z}I(z)\delta(x-x_0)\delta(y-y_0)$$

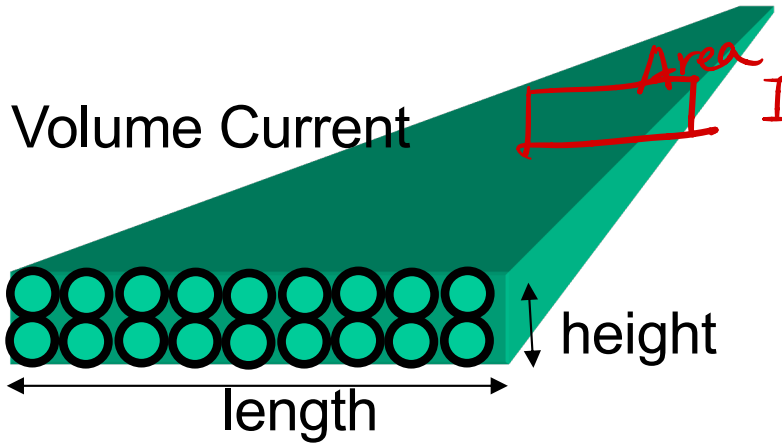
$[A/m^2]$ $[A]$ $[\frac{1}{m}]$ $[\frac{1}{m}]$



surface current density

$$\vec{J}(x,y,z) = \vec{J}_s(y,z)\delta(x-x_0)$$

$$\vec{J}_s \left[\frac{A}{m}\right]$$

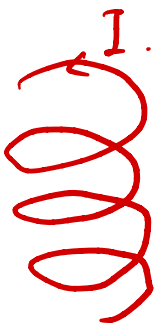


Area

$I_{enc} = J_v \cdot A = [\frac{A}{m^2}] \cdot [m^2] = [A]$

volumetric current density

$$\vec{J}_v \left[\frac{A}{m^2}\right]$$



$\oint \vec{D} \cdot d\vec{s} = Q_{enc}$

Challenge Question

- A coaxial cable has a solid center wire and an outer cylindrical shell wire. A uniform current density J flows in the direction $-\hat{x}$ on the inner wire and the total current returns on the outer shell. In which region(s) will the H-field be constant?

$J = \frac{I}{\pi a^2}$

$\oint_c \vec{H} \cdot d\vec{\ell} = I_{enc}$

$$H \cdot 2\pi r = \begin{cases} J \cdot \pi r^2 = \frac{I}{\pi a^2} \cdot \pi r^2 = \frac{I r^2}{a^2} & r<a \\ I & a<r<b \\ I - I = 0 & r>b \end{cases}$$

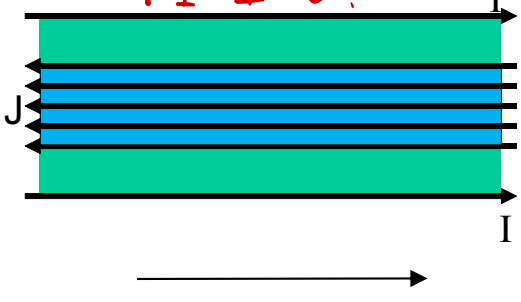
(a) r<a

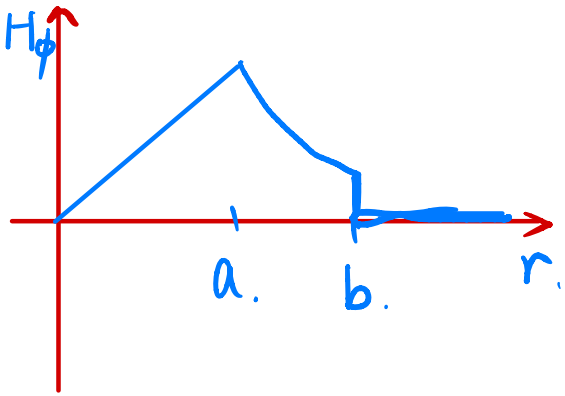
(b) a<r<b

(c) r>b

(d) both (b) and (c)







$$H_\phi = \begin{cases} \frac{Ir}{2\pi a^2} & r<a \\ \frac{I}{2\pi r} & a<r<b \\ 0 & r>b \end{cases}$$

Lecture 12 Summary

- Ampere's Force Law $d\vec{F}_1 = I_1 d\vec{l}_1 \times d\vec{B}_1$
- Biot-Savart Law $d\vec{B}_{at\,1} = \frac{\mu_o}{4\pi} \frac{I_2 d\vec{l}_2 \times \hat{a}_R}{R^2}$
- Infinite line of current $\vec{B} = \frac{\mu_o I}{2\pi r} \hat{a}_\phi$
- Lorentz Force Equation $\vec{F}_{TOTAL} = q\vec{E} + q\vec{v} \times \vec{B}$
- Magnetic Field Intensity $\vec{H} = \frac{\vec{B}}{\mu_o}$
- Next class
  - Ampere's Law